AFRICAN STANDARD

ARS/PAS 1261

First Edition 2025

Edible insects — Edible crickets — Specification

Reference No. ARS/PAS 1261:2025(E)
ICS 67.120.99

© ARSO 2025

Table of contents

1 Scope 1

2 Normative references 1

3 Terms and definitions 2

4 Requirements 3

4.1 General requirements 3

4.2 Specific requirements 3

5 Food additives 3

6 Hygiene 3

7 Contaminants 4

7.1 Heavy metal contaminants 4

7.2 Mycotoxins 4

7.3 Diseased insects, veterinary drugs and pesticide residues 4

8 Packaging and labelling 5

8.1 Packaging 5

8.2 Labelling 5

8.3 Allergen declarations 5

Annex A (informative) Inventory of cricket species in Africa 6

Annex B (informative) Specifications for *Acheta domesticus* (house cricket) products 7

Bibliography 8

 © ARSO 2025 — All rights reserved

Foreword

The African Organization for Standardization (ARS) is an African intergovernmental organization made up of the United Nations Economic Commission for Africa (UNECA) and the Organization of African Unity (AU). One of the fundamental mandates of ARSO is to develop and harmonize African Standards (ARS) for the purpose of enhancing Africa's internal trading capacity, increase Africa's product and service competitiveness globally and uplift the welfare of African communities. The work of preparing African Standards is normally carried out through ARSO technical committees. Each Member State interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, regional economic communities (RECs), governmental and non-governmental organizations, in liaison with ARSO, also take part in the work.

ARSO Standards are drafted in accordance with the rules given in the ISO/IEC Directives, Part 2.

The main task of technical committees is to prepare ARSO Standards. Draft ARSO Standards adopted by the technical committees are circulated to the member bodies for voting. Publication as an ARSO Standard requires approval by at least 75 % of the member bodies casting a vote.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ARSO shall not be held responsible for identifying any or all such patent rights.

This African Standard was prepared by the ARSO/PC 03, *Insects for food, feed and derived products*.

© African Organisation for Standardisation 2025 — All rights reserved*

ARSO Central Secretariat
International House 3rd Floor
P. O. Box 57363 — 00200 City Square
NAIROBI, KENYA

Tel. +254-20-2224561, +254-20-3311641, +254-20-3311608

E-mail: arso@arso-oran.org
Web: www.arso-oran.org

* © 2025 ARSO — All rights of exploitation reserved worldwide for African Member States' NSBs.

Copyright notice

This ARSO document is copyright-protected by ARSO. While the reproduction of this document by participants in the ARSO standards development process is permitted without prior permission from ARSO, neither this document nor any extract from it may be reproduced, stored or transmitted in any form for any other purpose without prior written permission from ARSO.

Requests for permission to reproduce this document for the purpose of selling it should be addressed as shown below or to ARSO's member body in the country of the requester:

© African Organisation for Standardisation 2025 — All rights reserved

ARSO Central Secretariat
International House 3rd Floor
P.O. Box 57363 — 00200 City Square
NAIROBI, KENYA

Tel: +254-20-2224561, +254-20-3311641, +254-20-3311608

E-mail: arso@arso-oran.org
Web: www.arso-oran.org

Reproduction for sales purposes may be subject to royalty payments or a licensing agreement. Violators may be prosecuted.

Introduction

One of the most consumed insects is cricket, which contributes to approximately 30% of the global market for edible insects [1]. The consumption of the crickets depends on their distribution and the cultural appropriateness of cricket consumption to people in a particular country [2]. The distribution of these edible crickets in five continents is as follows: Asia (41 species), Africa (26 species), America (five species), Europe (four species), and Australia (four species). Africa tops the list with 25 countries that consume various crickets, followed by Asia (13 countries), America (five countries), Europe (four countries), and Australia (two countries).

Globally, crickets are gaining recognition as a valuable alternative protein source for human consumption due to their lower resource requirement and ecological footprint compared to traditional livestock.

Globally, the sustainable use of crickets as food or feed is undermined by lack of information on the number of the edible crickets, the country where they are consumed, and the developmental stages consumed. Furthermore, lack of data on their nutritional content and the potential risks to potential consumers limits their consumption or inclusion into other food sources.

© ARSO 2025 — All rights reserved

Edible insects — Edible crickets — Specification

1 Scope

This African standard specifies the requirements, sampling and test methods for edible crickets processed and offered for human consumption.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

AOAC 984.27, *Calcium, copper, iron, magnesium, manganese, phosphorus, potassium, sodium, and zinc in infant formula — Inductively coupled plasma emission spectroscopic method*

AOAC 2001.04, *Determination of fumonisins in corn and corn flakes — Liquid chromatography with immunoaffinity column cleanup*

AOAC 2015.01, *Heavy Metals in Food — Inductively Coupled Plasma–Mass Spectrometry*

AOAC 2020.07, *Available carbohydrates in cereal and cereal products, dairy products, vegetables, fruit and food products, and animal feeds*

ARS 53, *General principles of food hygiene — Code of practice*

ARS 56, *Prepackaged foods — Labelling*

ARS/PAS 1292, *Edible insects — Good agricultural and wild collection practices*

ARS/PAS 1295, *Edible insects — Guide on good hygiene practices for insect production*

ARS 1713 *Nutrition labelling— Requirements*

ARS 1715, *Use of nutritional and health claims — Requirement*

CODEX STAN 192, *General standard for food additives*

CODEX STAN 193, *General standard for contaminants and toxins in products intended for human and animal consumption*

ISO 936, *Meat and meat products — Determination of total ash*

ISO 937, *Meat and meat products — Determination of nitrogen content — Reference method*

ISO 1442, *Meat and meat products — Determination of moisture content — Reference method*

ISO 1443, *Meat and meat products — Determination of total fat content*

ISO 4833-1, *Microbiology of the food chain — Horizontal method for the enumeration of microorganisms — Part 1: Colony count at 30 ºC by the pour plate technique*

ISO 5983-1, *Animal feeding stuffs — Determination of nitrogen content and calculation of crude protein content — Part 1: Kjeldahl method*

ISO 5985, *Animal feeding stuffs — Determination of ash insoluble in hydrochloric acid*

© ARSO 2025 — All rights reserved

ISO 6633, *Fruits, vegetables and derived products — Determination of lead content — Flameless atomic absorption spectrometric method*

ISO 6496, *Animal feeding stuffs — Determination of moisture and other volatile matter content*

ISO 6579, *Microbiology of food — Horizontal method for the detection of Salmonella spp*

ISO 6888-1, *Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of coagulase-positive staphylococci (staphylococcus Aureus and other soecies) — Part 1: Technique using baird-pac*

ISO 7251, *Microbiology of food — Horizontal method for the detection and enumeration of suspected Escherichia coli — Most probable number technique*

ISO 10272-1, *Microbiology of the food chain-Horizontal method for detection and enumeration of Campylobacter spp. — Part 1: Detection method*

ISO 11290-2, *Microbiology of the food chain-Horizontal method for the detection and enumeration of Listeria monocytogenes and of Listeria spp. — Part 2: Enumeration method*

ISO 13547-2, *Copper, lead, zinc and nickel sulphide concentrates — Determination of arsenic — Part 2: Acid digestion and inductively coupled plasma atomic emission spectrometric method*

ISO 16050, *Foodstuffs — Determination of aflatoxin B_1, and the total content of aflatoxins B_1, B_2, G_1 and G_2 in cereals, nuts and derived products — High-performance liquid chromatographic method*

ISO 16649-2, *Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of B-glucuronidase-positive Escherichia coli — Part 1: Colony-count technique at 44 using membranes and 5-bromo-4*

ISO 21527-1, *Food microbiology — Horizontal method for the enumeration of yeasts and moulds — Part 1: Colony count technique in products with water activity greater than 0.95*

ISO 21527-2, *Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of yeasts and moulds — Part 2: Colony count technique in products with water activity less than or equal to 0,95*

ISO 23776, *Meat and meat products — Determination of total phosphorous content*

ISO 27085, *Animal feeding stuffs — Determination of calcium, sodium, phosphorus, magnesium, potassium, iron, zinc, copper, manganese, cobalt, molybdenum, arsenic, lead and cadmium by ICP-AES*

3 Terms and definitions

For the purposes of this standard, and the following terms and definitions apply.

3.1
edible insect product
product that has been processed from edible insects harvested either from domesticated farms or from the wild. The product may be presented in the form of whole insect, milled, paste, liquid/aqueous or in powder form

3.2
whole insect product
edible insect product either whole or where some inedible parts of the insects such as wings, gut hair have been removed during or after drying

 © ARSO 2025 — All rights reserved

3.3
powdered insect product
particulate product of dried insects that is achieved by milling or grinding dry whole insect product

4 Requirements

4.1 General requirements

4.1.1 Edible crickets intended for human consumption shall be of recognized species which are proven to be safe for human consumption. A list of edible crickets is provided in Annex A.

4.1.2 Edible crickets to processed into edible cricket products shall be harvested from the wild in accordance with ARS 1292 or farmed in accordance with ARS/PAS 1294. The processing shall comply with ARS/PAS 1295.

4.1.3 Edible crickets shall be:

a) Presented in form of either (i) blanched whole dried; (ii) roasted / fried whole; (iii) cricket protein granules / powder or paste; (iv) cricket dukkha

b) free from adulterants, foreign matter and objectionable odour; and

c) free from infestation and contamination from pests

4.1.4 Cricket products may contain other food ingredients to enhance flavour and palatability such as edible salt, seasonings and herbs complying with relevant African Standards.

4.2 Specific requirements

Cricket products shall conform to the specific requirements when tested with the test methods described in table:

Table 1 — Specific requirements for cricket products

S/N	Parameter	Limits, %,	Test method
i)	Moisture content, m/m, max	8	ISO 6496
ii)	Protein content % m/m min.	45	ISO 5983-1
iii)	Fat content, % m/m min	10	ISO 1443
iv)	Total ash, %, m/m min	5	ISO 936
iii)	Acid insoluble ash, (max), %, m/m	1	ISO 5985

5 Food additives

Only permitted food additives in accordance with CODEX STAN 192 for similar products may be used.

6 Hygiene

6.1 The cricket products shall be prepared and packaged in premises built and maintained under hygienic condition in accordance with ARS 53 and ARS/PAS 1295.

6.2 Cricket products shall be free from any pathogenic microorganism and shall in particular comply with the microbiological requirements indicated in Table 2 when tested in accordance with the test methods specified therein.

© ARSO 2025 — All rights reserved

Table 2 — Microbiological limits for dried Insect products

S/N	Type of micro-organism	Maximum number of counts	Method of test
i	Total Aerobic Counts at 30 °C, CFU/g	10^5	ISO 4833-1
ii	*Salmonella* in 25 g	Not detected	ISO 6579-1
iii	Escherichia coli, CFU/g	Less than 10[a]	ISO 16649-2
iv	*Staphylococci aureas* CFU/g	Less than $10/10^2$	ISO 6888-1/2
v	*Listeria monocytogenes,* CFU/25g	10^2	ISO 11290-2
vi	*Cronobacter spp.(Enterobacter sakazakii) CFU/10g*	Absent	ISO 22964
vii	*Campylobacter, CFU/25g*	Absent	ISO 10272-1
viii	Yeast and moulds CFU/g	10^2	ISO 21527-2
[a] For ready to use. Products that will require further processing before consumption as per the direction of use a maximum of 10^2 shall apply.			

7 Contaminants

7.1 Heavy metal contaminants

Edible cricket products shall not contain heavy metal contaminants in amounts established to cause adverse effect to health and in particular shall comply with the limits indicated in Table 3 when tested in accordance with the test methods specified therein.

Table 3 — Limits for heavy metal contaminants for edible cricket products

S/N	Contaminant	Max. limit, mg/kg	Test method
i)	Lead (Pb)	0.1/0.5	ISO 6633
iii)	Cadmium (Cd)	0.03/0.5	ISO 27085

7.2 Mycotoxins

The maximum content of aflatoxins in edible crickets when determined in accordance with the method described in ISO 16050 shall not exceed 5 µg/kg for aflatoxin B_1 and 10 µg/kg for total aflatoxins.

7.3 Diseased insects, veterinary drugs and pesticide residues

7.3.1 Veterinary drugs and hormones

ARS 1296 provides guidance on the use of veterinary drugs, hormones and pesticides in rearing insects. Veterinary drugs may be sprayed or mixed with substrates during rearing of insects to combat infections with a variety of insect pathogens, and this use may also result in residues in the insects. In addition, other substances, such as hormones, if used in farming could result in residues [3].

7.3.2 Pesticide residues

7.3.2.1 Substrates for insect rearing may contain residues of pesticides. In terms of feed safety, the Codex Alimentarius recommends that the concentrations of chlorpyrifos and piperonyl butoxide in animal feed (alfalfa and pea fodder, respectively) should be below 5 000 µg/kg and 2 000 µg/kg, respectively.

 © ARSO 2025 — All rights reserved

7.3.2.2 Make sure that the insects that you are collecting, and their host plant, have not been sprayed with pesticides [4]. This can be the case when collecting locusts or grasshoppers in agricultural fields or palm weevil larvae on palm plantations. After spraying, the pesticides may remain in the insect and can be toxic to humans.

7.3.3 Dead insects: Do not collect insects found already dead.

8 Packaging and labelling

8.1 Packaging

8.1.1 The edible cricket products shall be packaged in food grade materials that will ensure the safety and integrity of the product throughout the shelf life.

8.1.2 The fill of the container shall be in accordance with legal metrology requirements.

8.2 Labelling

8.2.1 In addition to the labelling requirements of ARS 56, the label shall include the following information:

a) Name of the product shall be either ‘Dried whole crickets’ or ‘Dried powdered/ paste crickets’ or ‘Dried milled crickets’ or ‘Fried cricket products’ or ‘Grinded/ground cricket products’;

b) Declare the insect by common name and biological name in brackets;

c) Direction for use of the product;

d) List of ingredients

e) Name and address of the manufacturer

f) Net weight of the contents

g) Storage instructions.

h) production date;

i) batch number;

j) Best before date; and

k) country of origin.

8.2.2 Any health and/or nutrition claim on dried insect products shall conform to the provisions of ARS 1713 and ARS 1715.

8.3 Allergen declarations

Crickets may cause allergenic reactions in people who are sensitive to shellfish and crustaceans.

© ARSO 2025 — All rights reserved

Annex A
(informative)

Inventory of cricket species in Africa

Family	Species name	Common name	Country	Stage consumed
Gryllidae	*Acheta domesticus* (Lin.)	House cricket	Worldwide	Adult
Gryllidae	Acheta spp.	Cricket	Zambia, Zimbabwe, Kenya	Adult
Gryllidae	*Brachytrupes membranaceus* (Drury)	Tobacco cricket/ Giant/Large African cricket	Republic of Congo, Burkina Faso, Tanzania, Angola, Togo, Benin; Malawi	Adult; Nymph
Gryllidae	*Gryllus similis* (Chapman)	Field cricket	Nigeria; Ghana	Adult
Gryllidae	*Gryllus thusilis*	Jamaican field cricket		Adult
Gryllidae	*Gryllodes sigillatus*	Banded cricket/ Tropical house cricket	Tropical Africa	Adult
Gryllidae	*Gryllus bimaculatus*	African/ Mediterranean field cricket	Sierra Leone, Liberia, Benin, Democratic Republic of Congo, Kenya, Sudan, Zambia	Adult
Gryllidae	*Scapsipedus icipe* Hugel & Tanga	Field cricket	Kenya	Adult
Gryllotalpidae	*Gryllotalpa africana*	Mole cricket	Nigeria; Cameroon; Zimbabwe; Malawi; Kenya; Tanzania; DR Congo; Uganda; South Africa	Adult
Gryllotalpidae	*Gryllotalpa gryllotalpa*	European mole cricket	Malawi	Adult
Gryllotalpidae	*Scapteriscus vicinus* (Scudder)	Ground cricket	Ghana	Adult
Stenopelmatidae	*Henicus whellani* (Chopard)	Ground-dwelling cricket	Zimbabwe	Adult
Tettigoniidae	*Acanthoplus discoidalis*	Bush cricket/ armoured ground cricket	Angola; Namibia; Botswana; Zimbabwe; South Africa	Adult
Gryllidae	*Gymnogryllus lucens*	Cricket	Nigeria	Adult
Gryllidae	Gryllus campestris	European field cricket	Burkina Faso; Cameroon, Malawi	Adult
Conocephalidae	*Homorocoryphus vicinus*	Cricket	Uganda	Adult
Tettigoniidae	*Homorocoryphus nitidulus*	Bush Cricket	Cameroon	Larvae
Gryllidae	*Brachytrupes spp.*	Mole crickets, Tobacco Cricket	Uganda, Cameroon	Adult

© ARSO 2025 — All rights reserved

Annex B
(informative)

Specifications for *Acheta domesticus* (house cricket) products

Characteristics	Partially defatted powder [a] [5]	AD frozen or paste [6]	AD dried or AD powder [6]
Crude protein (N x 6.25), % w/w	74.0 – 78.0	12 – 21	55 – 65
Fat, % w/w	9.0 – 12.0	3 – 12	29 – 35
Saturated fat, % w/w	3.5 – 5.0	36 – 45	36 – 45
Moisture, % w/w	3.0 – 6.0	76 – 82	≤ 5
Dietary fibre, % w/w	8.0 – 10.0	0.8 – 3	3 – 6
Chitin[b], % w/w	4.0 – 8.5	≤ 3	≤ 10
Ash, % w/w	≤ 5.6	≤ 1.5	≤ 5
Digestible carbohydrates, % w/w		0.1 – 2	1.0 – 4.0
Peroxide value (Meq O2/kg fat	≤ 5.0	≤ 5	≤ 5
Manganese, mg/kg	≤ 100.0		
Cyanide, mg/kg	≤ 5.0		
Heavy metals			
Lead, mg/kg	≤ 0.1	≤ 0.05	≤ 0.05
Cadmium, mg/kg	≤ 0.025	≤ 0.06	≤ 0.06
Mycotoxins			
Aflatoxins (B_1+B_2+G_1+G_2), µg/kg	≤ 0.4	≤ 4.0	≤ 4.0
Aflatoxin B_1, µg/kg		≤ 2	≤ 2
Deoxynivalenol, µg/kg	≤ 10	≤ 200	≤ 200
Ochratoxin A, µg/kg	≤ 0.5	≤ 0.4	≤ 0.4
Sum of dioxins and dioxins-like PCBs (UB WHO_{2005} PCDD/F-PCBTEQ), pg/g fat	≤ 1.25	≤ 1.25	≤ 1.25
Microbiological			
Total aerobic microbial count, CFU/g	< 10^5	≤ 10^5	≤ 10^5
Enterobacteriaceae (presumptive), CFU/g	≤ 100	≤ 100	≤ 100
Escherichia coli, CFU/g	≤ 50	≤ 50	≤ 50
Listeria monocytogenes, in 25 g	Not detected	Not detected	Not detected
Salmonella spp., in 25 g	Not detected	Not detected	Not detected
Bacillus cereus (presumptive), CFU/g	≤ 100	≤ 100	≤ 100
Coagulase-positive staphylococci, CFU/g	≤ 100	≤ 100	≤ 100
Sulfite-reducing anaerobes, CFU/g		≤ 30	≤ 30
Total yeast and mould count, CFU/g	≤ 100	≤ 100	≤ 100
AD frozen: thermally processed, frozen, whole *A. domesticus* AD dried: thermally processed, dried, whole *A. domesticus* AD powder: thermally processed, dried, whole, ground *A. domesticus*			

[a] The novel food is partially defatted powder obtained from the whole *Acheta domesticus* (house cricket) following a series of steps involving a 24 fasting period of the insects to allow them discard their bowel content, the sacrifice of the insects by freezing, washing, thermal processing, drying, oil extraction (mechanical extrusion), and grinding.

[b] Chitin calculated as the difference between the Acid Detergent Fibre fraction and the Acid Detergent Lignin fraction (ADFADL), as described in [7]

[c] UB: upper bound; WHOPCDD/ F-PBC-TEQ: sum of polychlorinated dibenzo-p-dioxins, polychlorinated dibenzofurans, polychlorinated biphenyls expressed as World Health Organization toxic equivalent factors (TEF) set by the World Health Organization in 2005

© ARSO 2025 — All rights reserved

Bibliography

[1] Kemsawasd, V., Inthachat, W., Suttisansanee, U. and Temviriyanukul, P. (2022) Road to The Red Carpet of Edible Crickets through Integration into the Human Food Chain with Biofunctions and Sustainability: A Review. *International Journal of Molecular Sciences*, Multidisciplinary Digital Publishing Institute. **23**, 1801. https://doi.org/10.3390/ijms23031801

[2] Magara, H.J.O., Niassy, S., Ayieko, M.A., Mukundamago, M., Egonyu, J.P., Tanga, C.M. et al. (2021) Edible Crickets (Orthoptera) Around the World: Distribution, Nutritional Value, and Other Benefits—A Review. *Frontiers in Nutrition*, **7**.

[3] Belluco, S., Losasso, C., Maggioletti, M., Alonzi, C.C., Paoletti, M.G. and Ricci, A. (2013) Edible Insects in a Food Safety and Nutritional Perspective: A Critical Review. *Comprehensive Reviews in Food Science and Food Safety*, **12**, 296–313. https://doi.org/10.1111/1541-4337.12014

[4] Cloutier, J. (2015) Edible Insects in Africa - An Introduction to Finding, Using and Eating Insects. 1st Edition. Technical Centre for Agricultural and Rural Cooperation (CTA), Wageningen, The Netherlands.

[5] EFSA Panel on Nutrition, Novel Foods and Food Allergens (NDA), Turck, D., Bohn, T., Castenmiller, J., De Henauw, S., Hirsch-Ernst, K.I. et al. (2022) Safety of Partially Defatted House Cricket (Acheta domesticus) Powder as a Novel Food Pursuant to Regulation (EU) 2015/2283. *EFSA Journal*, **20**. https://doi.org/10.2903/j.efsa.2022.7258

[6] EFSA Panel on Nutrition, Novel Foods and Food Allergens (NDA), Turck, D., Bohn, T., Castenmiller, J., De Henauw, S., Hirsch-Ernst, K.I. et al. (2021) Safety of Frozen and Dried Formulations from Whole House Crickets (Acheta domesticus) as a Novel Food Pursuant to Regulation (EU) 2015/2283. *EFSA Journal*, **19**, e06779. https://doi.org/10.2903/j.efsa.2021.6779

[7] Hahn, T., Roth, A., Febel, E., Fijalkowska, M., Schmitt, E., Arsiwalla, T. et al. (2018) New Methods for High-Accuracy Insect Chitin Measurement. *Journal of the Science of Food and Agriculture*, **98**, 5069–73. https://doi.org/10.1002/jsfa.9044

 © ARSO 2025 — All rights reserved

© ARSO 2025 — All rights reserved